University of Dhaka
Department of Computer Science and Engineering

Project Report:
Fundamentals of Programming Lab(CSE-1211)

Project Name:
Billiard++

Team Members:
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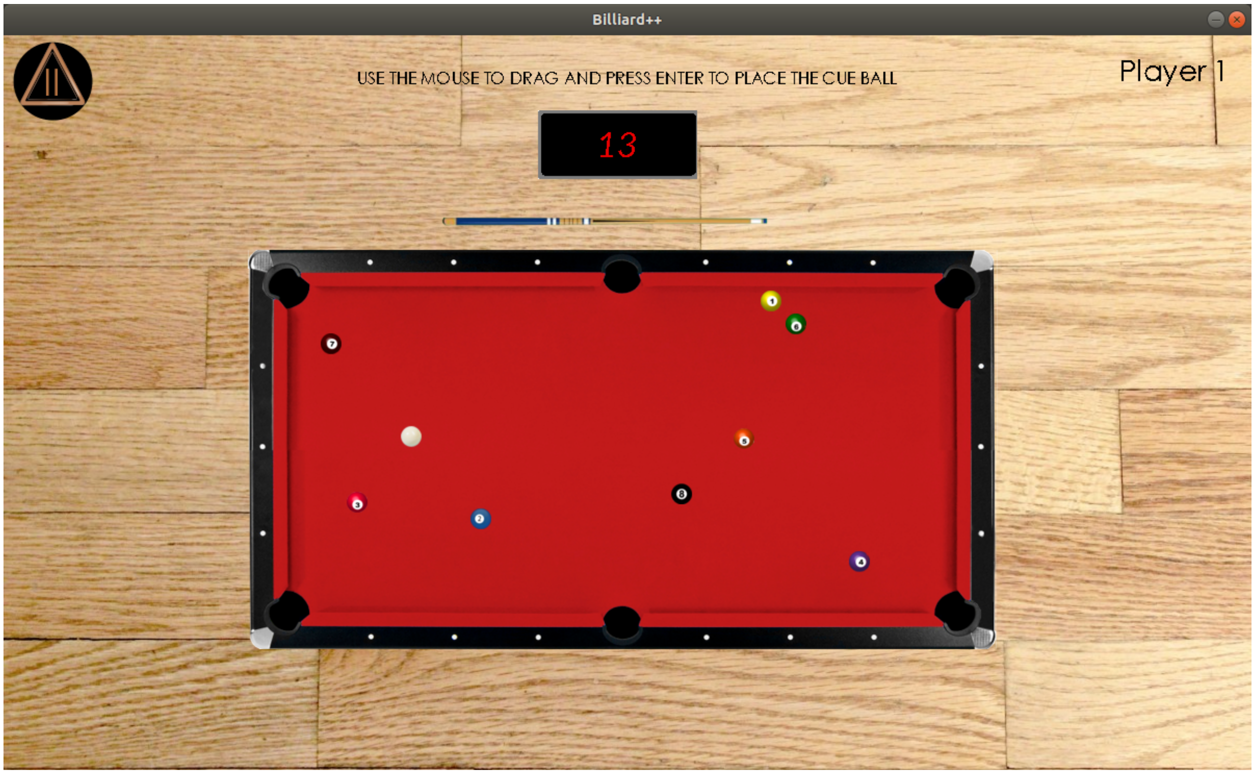
Hasan Nur Towha

1. **Introduction:**
   “Billiard++” is a cue sports game that tries to refine some of the existing billiard games having added some unique features of its own. The game features 4 different but interesting game modes. Among them, the two game modes, “8 Ball Pool” and “9 Ball Pool” are widely popular while the other two modes, “Replication Mode” and “Beat The Clock” add some individuality to the game. The game also features a completely new and intriguing UI as well as a convenient login utility.
2. **Objectives:**
   The main objective of every game is to entertain its user and provide them with a quality experience. “Billiard++” is no exception to this. The two game modes, “8 Ball Pool” and “9 Ball Pool”, provides the user with a chance to have a face-off with a friend while the two other modes, “Replication Mode” and “Beat the Clock” gives the user a challenging experience that they have to overcome with their own skills. The latter two modes very well serve the purpose of passing quality time amidst boredom.
3. **Project Features:**
   The signature features of this project are:
   a. “Replication” Mode
   b. “Beat the Clock”
   c. Unique User Login Utility

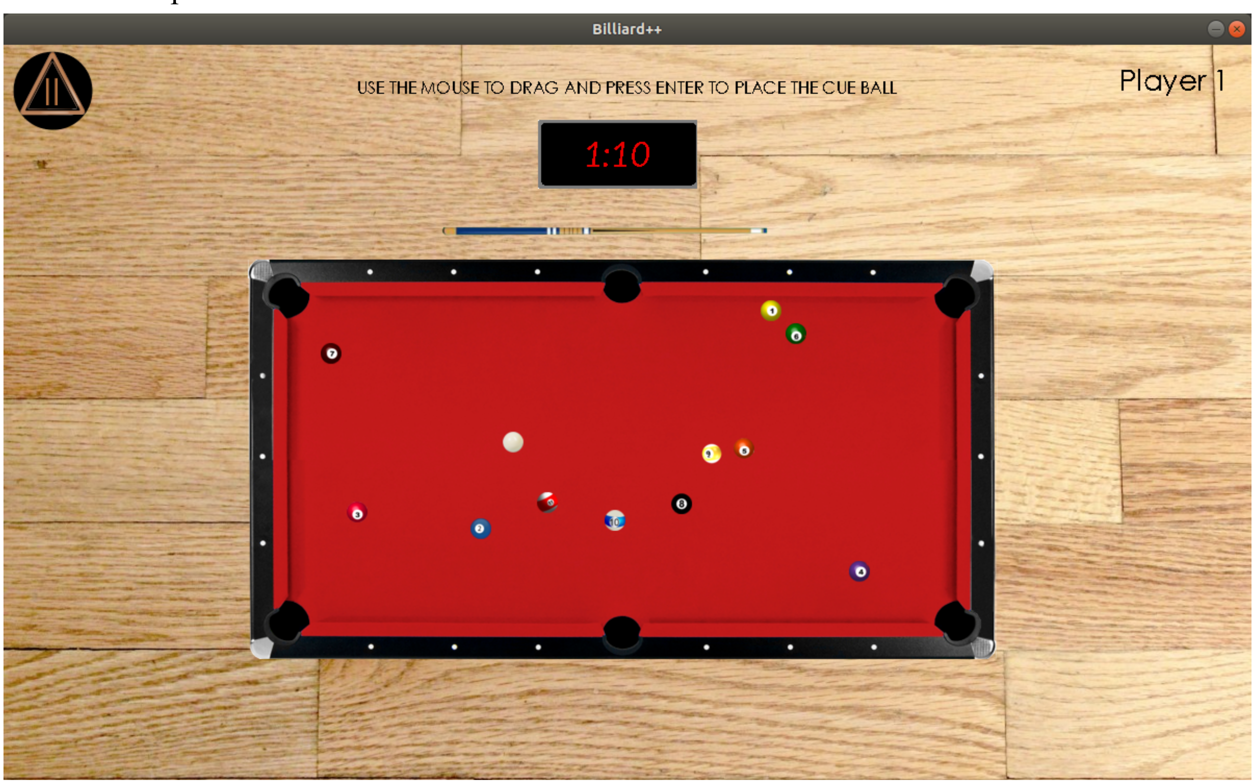
   Apart from these, the game also features:
   a. “8 Ball Pool” Mode (For 2 players)
   b. “9 Ball Pool” Mode (For 2 players)

The “Replication” Mode is an all-new game mode that involves replication of the available balls anytime the user fails to pocket a ball or commits foul.

The following is a screenshot of the mode at the beginning of a round:



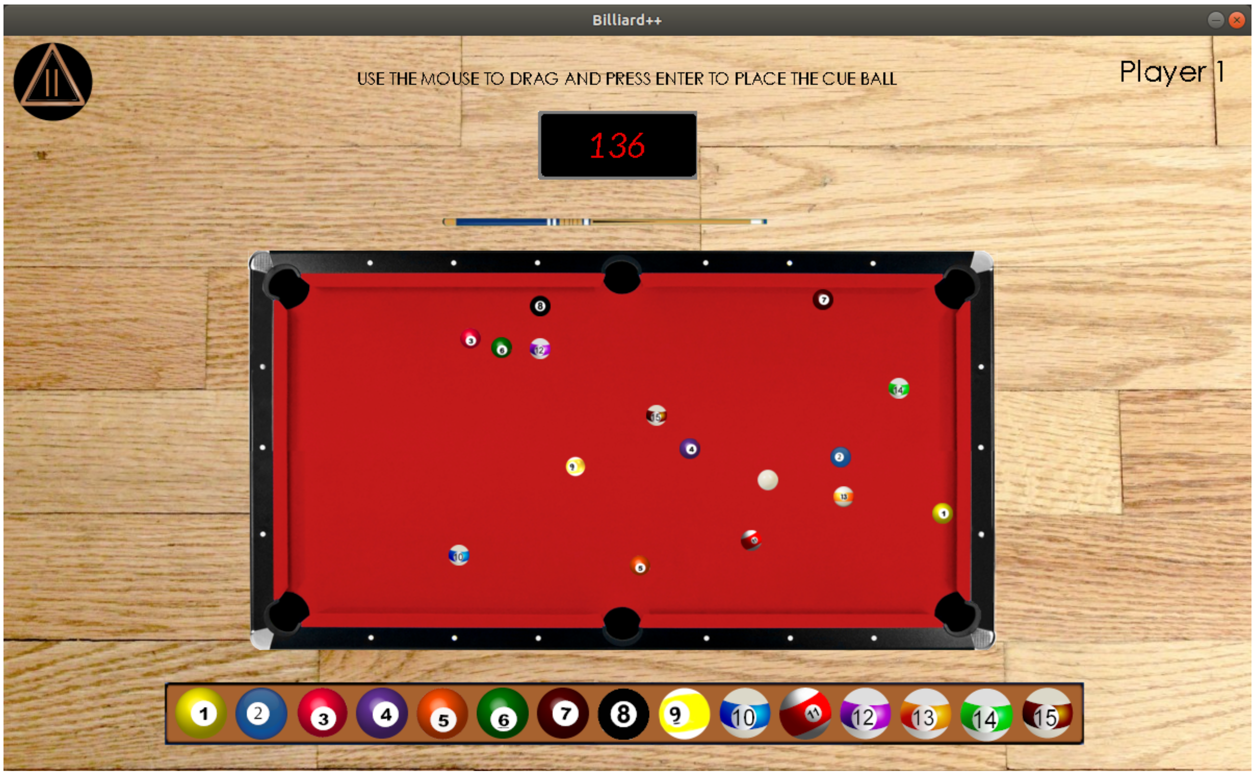
When the user commits a foul, three balls are generated at random locations on the board, as shown in the picture below:



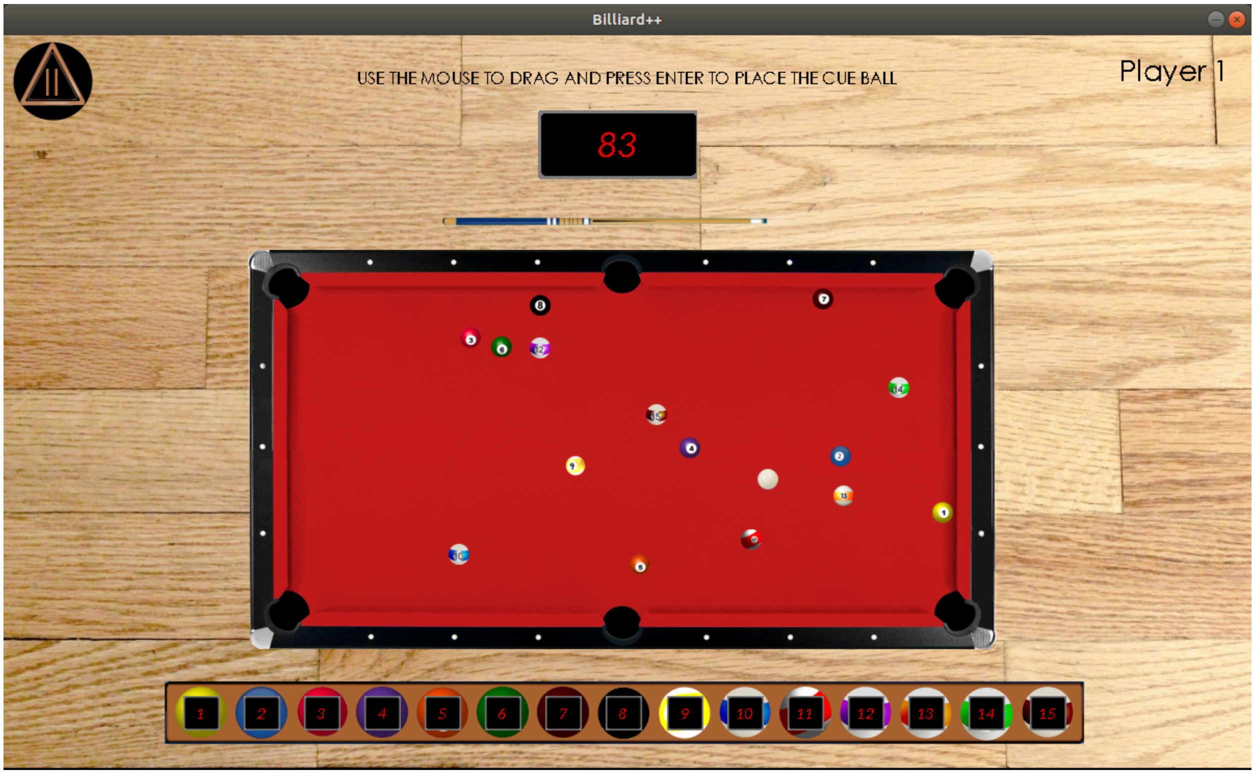
If the user fails to pocket at least one ball after performing a hit, then 2 balls are generated instead of 3.

The “Beat the Clock” is an innovative game mode that is based on the idea of regarding the user for the success and punishing them for failure.

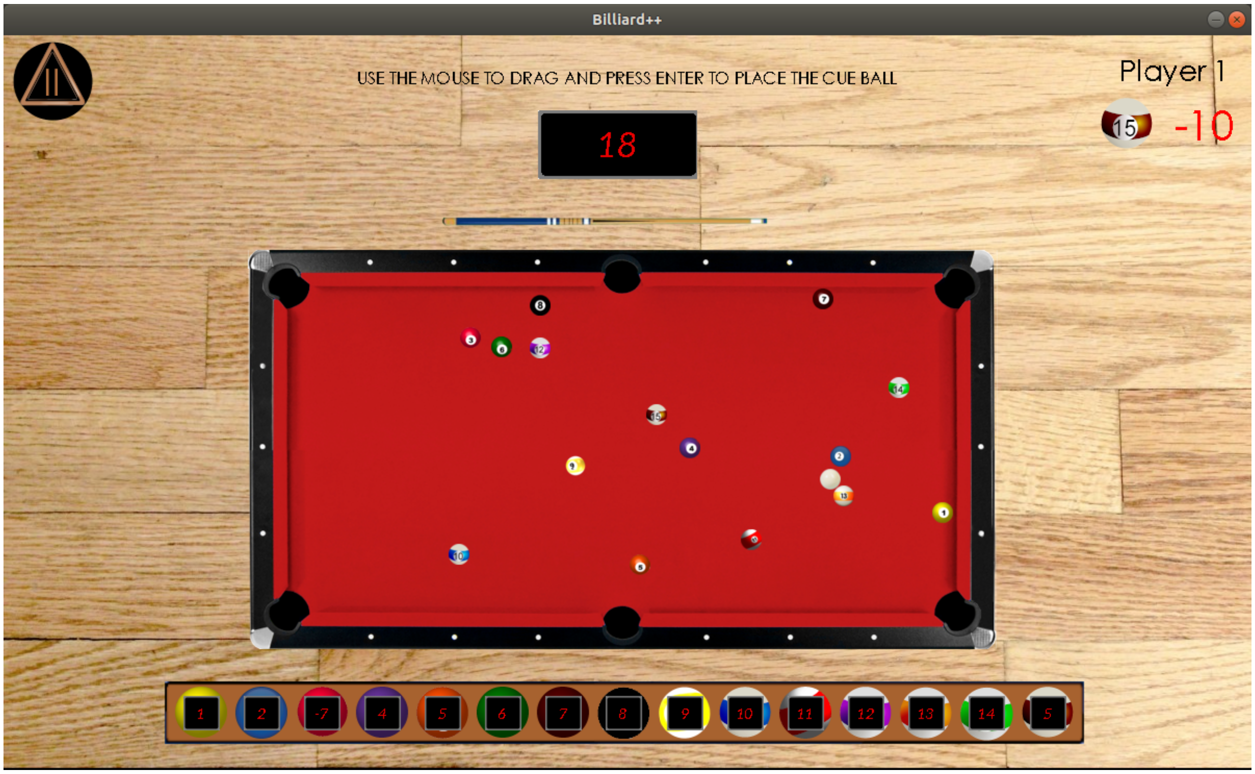
The following is a screenshot of the mode at the beginning of a round:



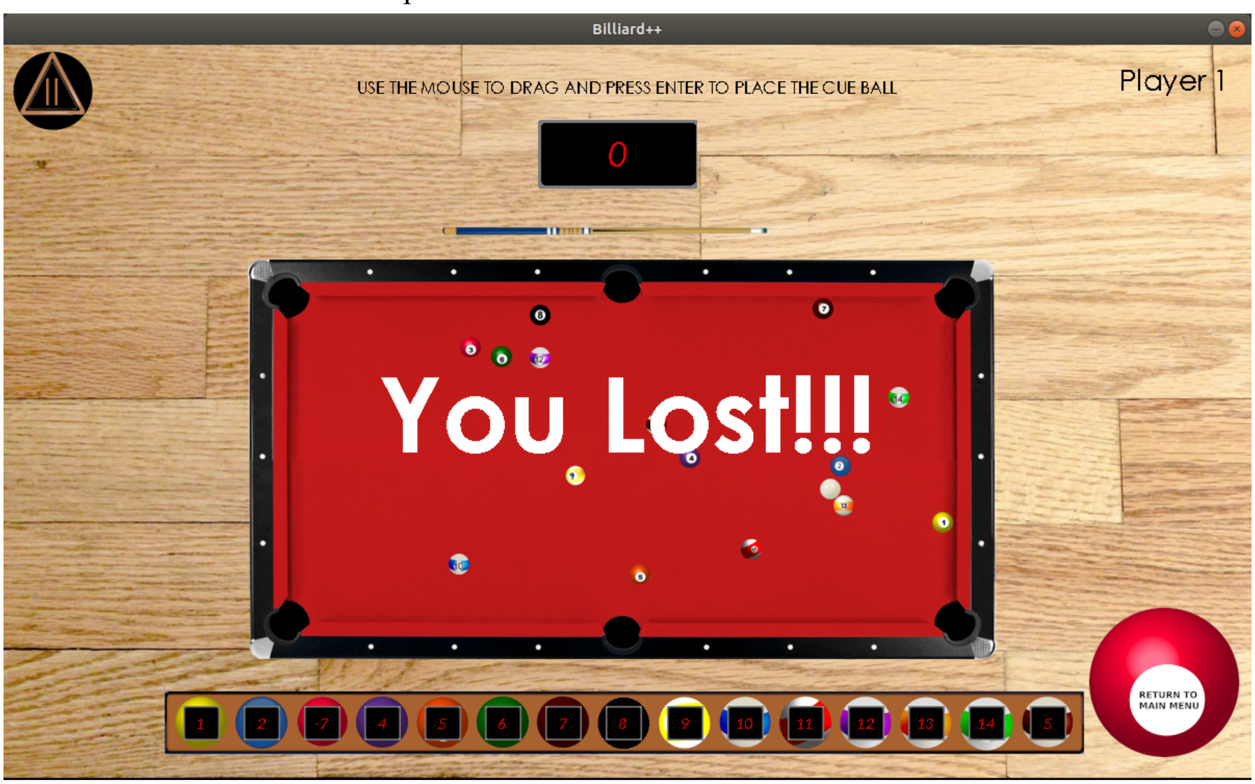
Each ball has a certain value at the beginning, which can be checked during the game as shown below:



Pocketing any particular ball adds the value of the ball to the timer.
The value of each ball also changes depending on the player’s performance. If the player performs well, then the value of any randomly selected ball is increased, giving the user more time as reward for pocketing that ball.
However, if the user commits a foul or fails to pocket a ball in 3 consecutive attempts, then the value of any randomly selected ball decreases as a punishment. This is shown in the following picture:

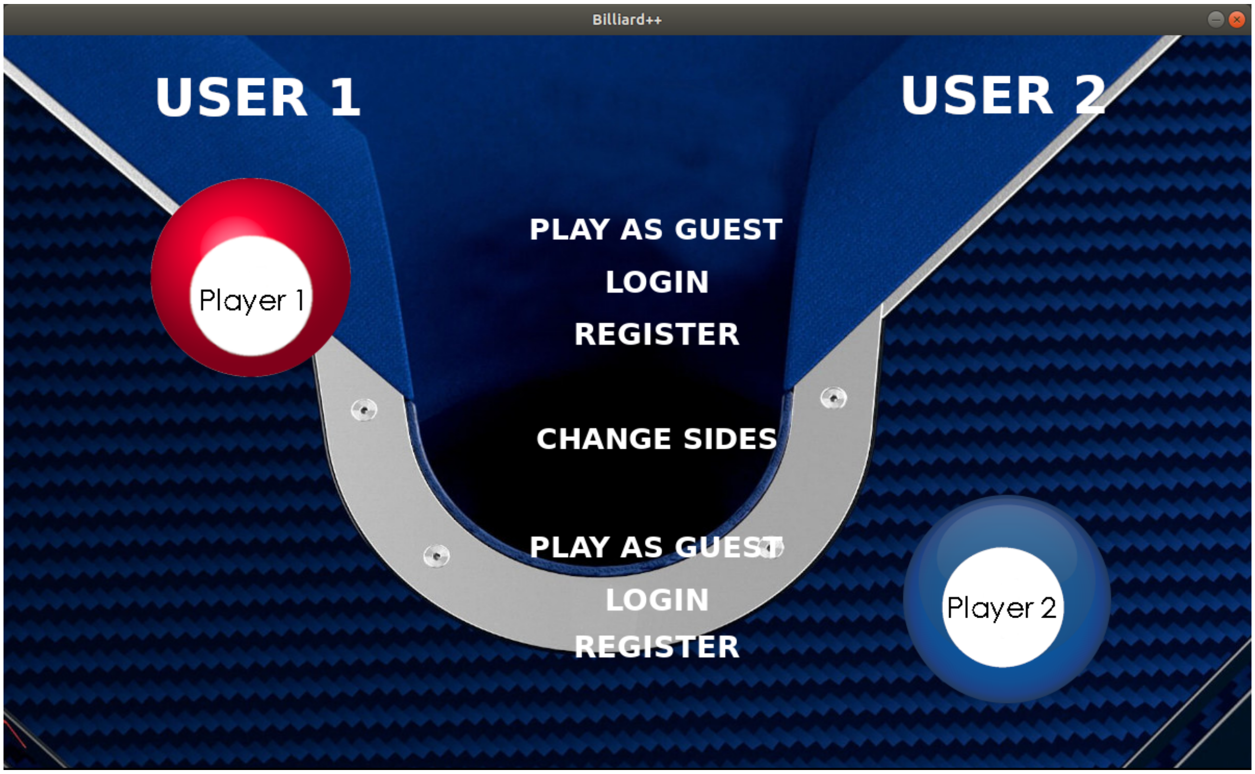


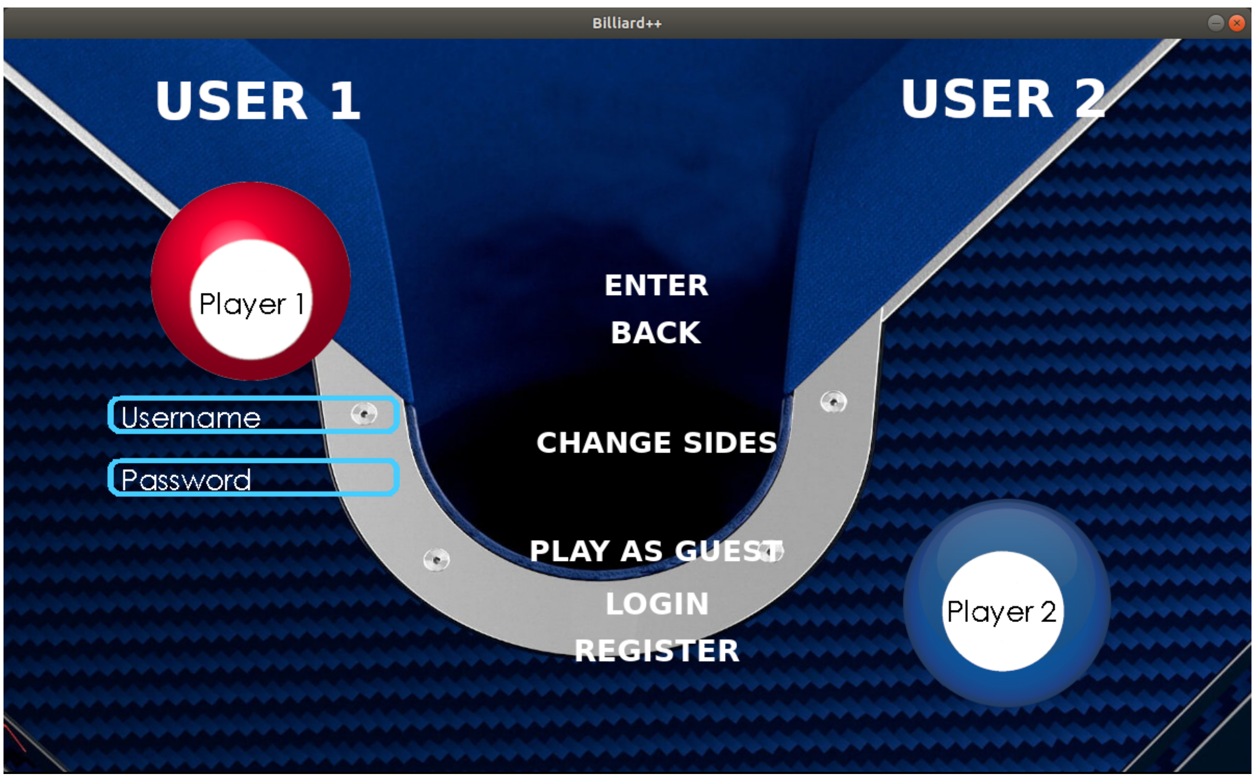
The user loses if he/she fails to pocket all the balls before the timer runs out.

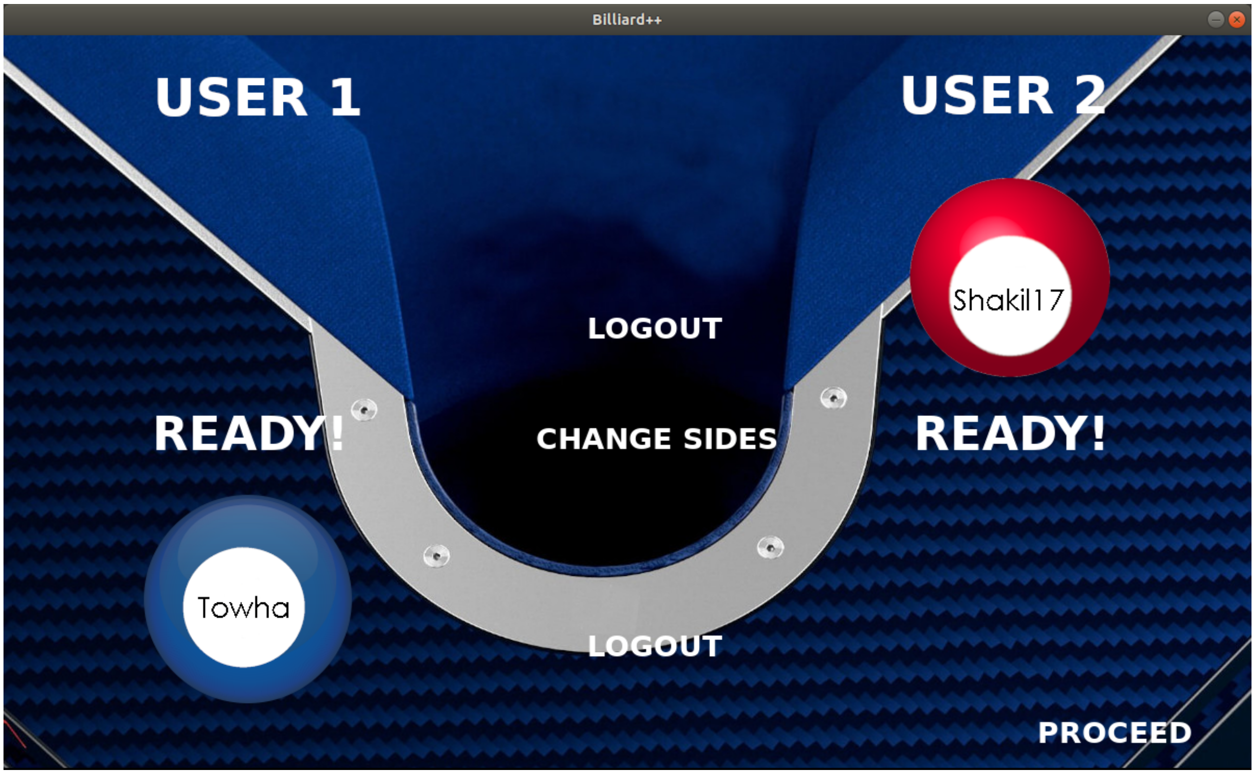


A detailed description of all the game modes is present in the Help section of the game.

The game also features a unique login system, where the user can register new ids, log into the existing ones or play as a guest. It also allows the user to switch sides in case they want to do so.

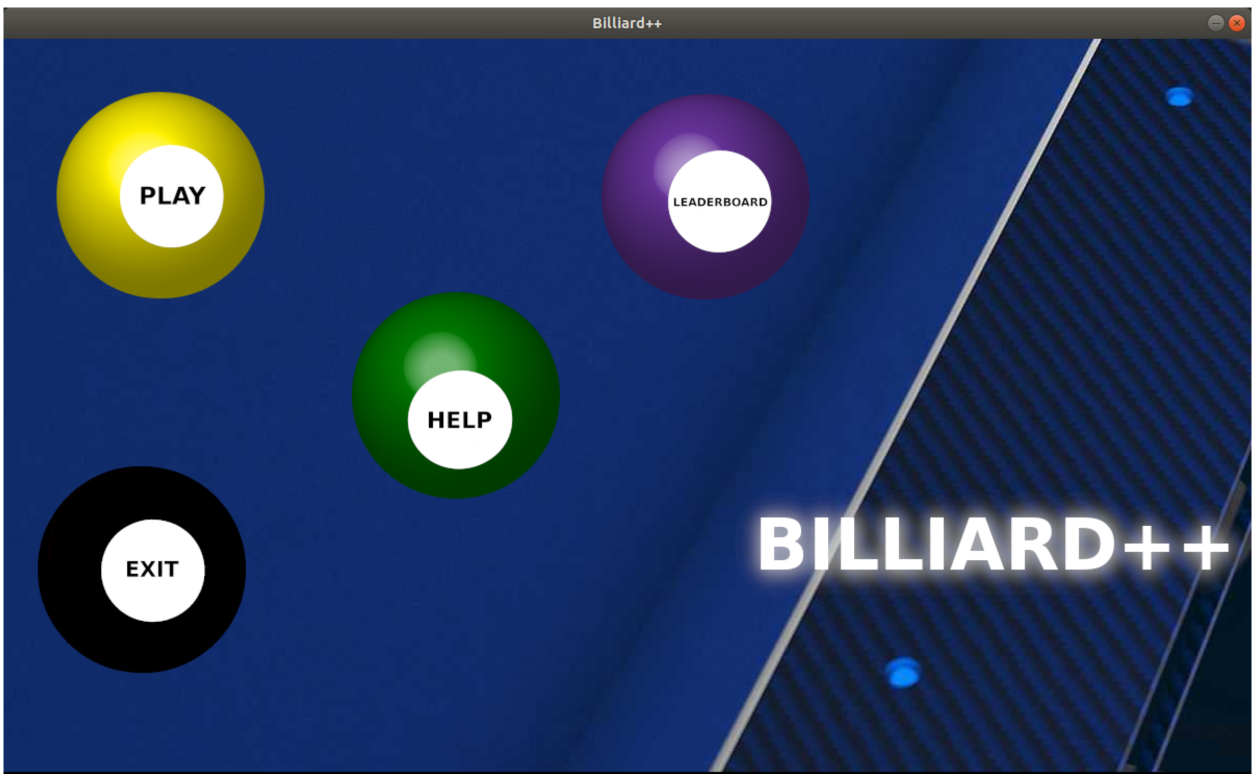


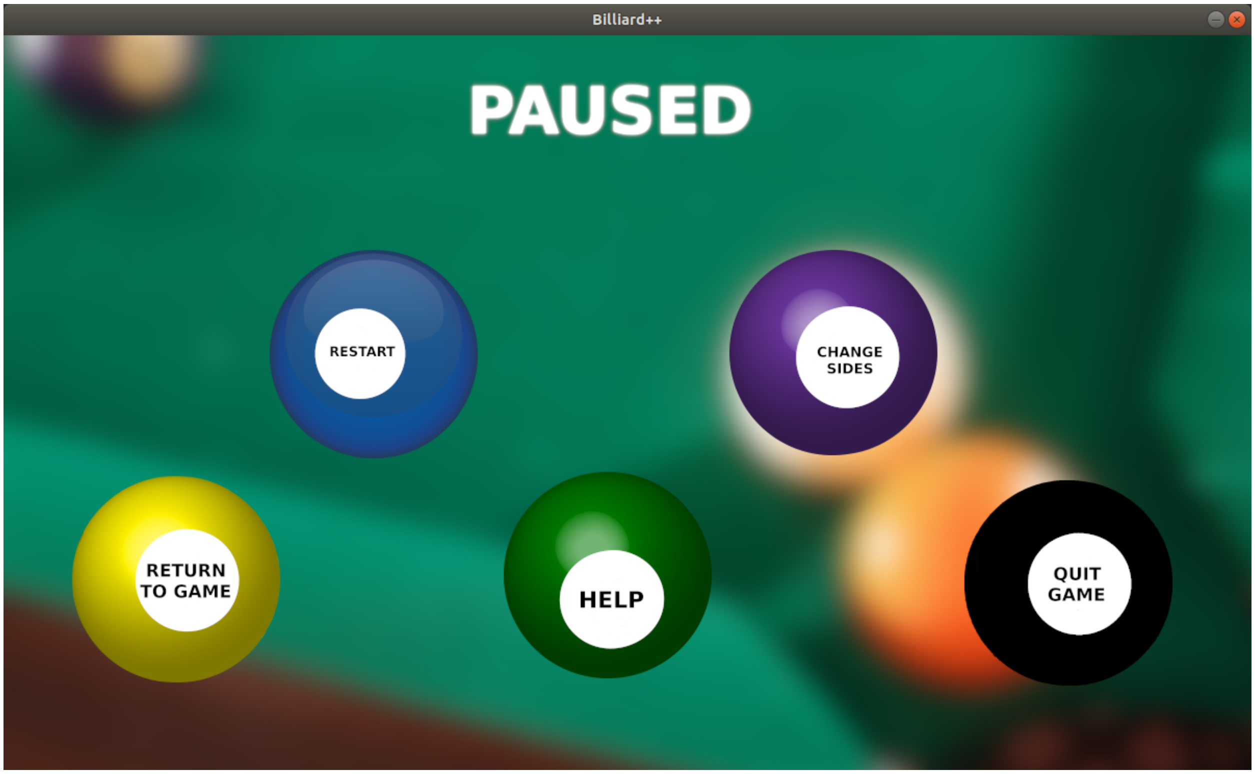


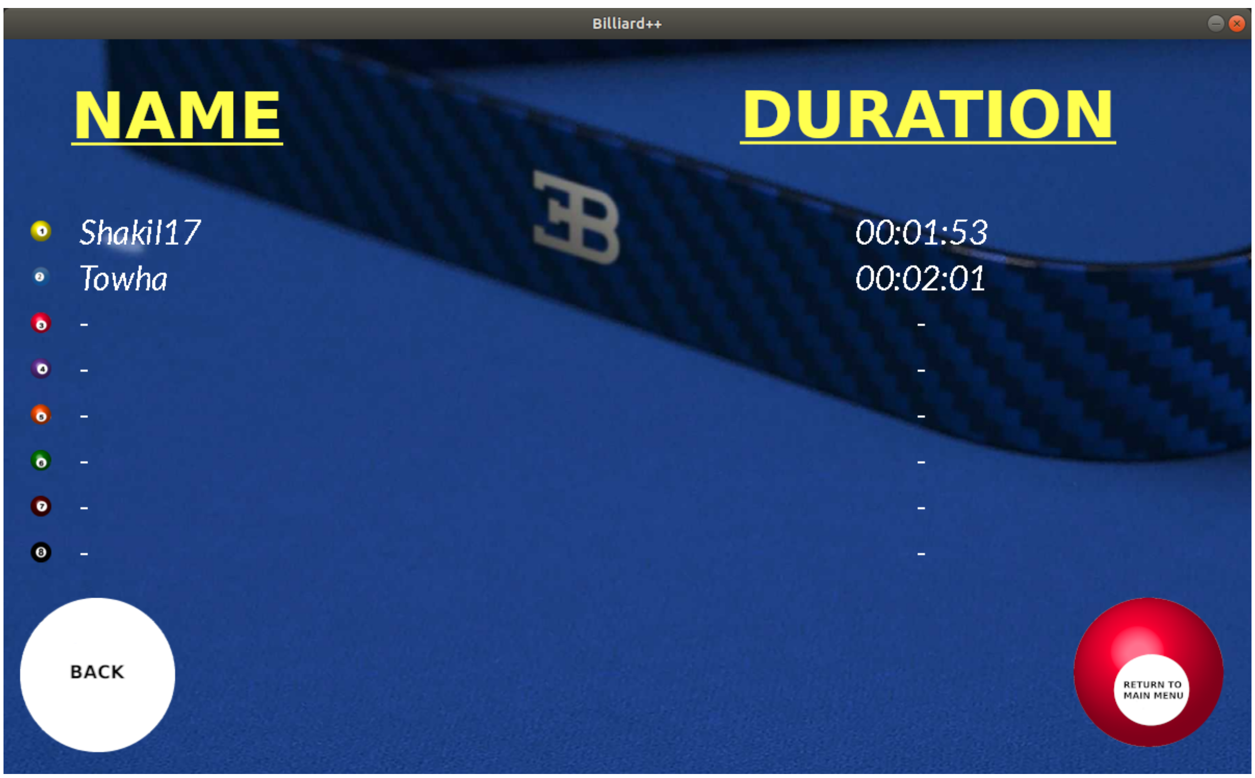


The game has “8 Ball Pool” and “9 Ball Pool” modes that can be played between 2 players.

Goes without saying, the game also contains a pause menu, leaderboards and proper help guidelines. The game overall is backed up by an eye-catching UI.







4. Project Modules:

The following header files were implemented:

- 8ball.h : This module contains necessary functions and variables for "8 Ball Pool" mode. It controls this mode of the game.
- 9ball.h : This module contains necessary functions and variables for "9 Ball Pool" mode. It controls this mode of the game.
- aimline.h: This module controls the virtual dash line to show the movement direction of the cue ball before it is hit, as well as the direction of movement of the ball to be hit by the cue ball confined to a constant scale.
- ball.h: This module controls the state and events of all the balls. Also contains many necessary info and variables of the overall game.
- btc_mode.h: This module contains necessary functions and variables for "Beat the Clock" mode. It controls this mode of the game.
- button.h: This module controls the circular and rectangular buttons. Button struct is for ball buttons and Rect_button struct controls rectangular buttons in the login menu.
- circle.h: This module contains functions about circle.
- consts.h: This module contains necessary constants.
- controls_menu.h: This module controls every page of the "controls" section.
- cue.h: This module controls the state and events of the cue.
- cue_hover.h: This module controls the highlight with a cue picture below the name of the current player in two player modes.
- cue_trigger.h: This module controls the triggering of cue for a shot.
- game_end_text.h: This module controls the rendering of text when result happens in single player modes.
- game_state.h: This module controls various in-game states like balls movement and merges the required effects. Also renders the winner declaration text in two player modes.
- help_menu.h: This module controls the "Help" section of the game and directs to rules,introduction and controls.
- introduction_menu.h: This module controls the pages of "Introduction" section of the game.
- leaderboard.h: This module controls the "Leaderboards" section of the game and directs to the leaderboards section of two single player modes.
- leaderboard_btc.h: This module controls the leaderboard of "Beat the Clock" mode. It also reads and updates leaderboard_btc.bin.
- leaderboard_replication.h: This module controls the leaderboard of "Replication" mode. It also reads and updates leaderboard_replication.bin.
- login_menu.h: This module controls the login,register and change sides system of the game .
- main_menu.h: This module controls the main menu of the game and directs to play,help,leaderboard and exit button.

- maths.h: This module contains necessary mathematical functions used in the project.
- modified_ball.h: This module controls the pipe and bigger balls and their states and also the toggling status board in "Beat the Clock" mode.
- pause_menu.h: This module controls the pause menu of all the modes.
- pipes.h: This module controls the pipes which show the pocketed balls in proper order.
- play_menu.h: This module controls the play section and directs to all the four modes of games.
- record.h: This module controls the records of the leaderboard of single player modes
- replication_mode.h: This module contains necessary functions and variables for "Replication" mode. It controls this mode of the game.
- rules_8ball.h: This module controls the rendering of all pages of the rules section of "8 Ball Pool" mode.
- rules_9ball.h: This module controls the rendering of all pages of the rules section of "9 Ball Pool" mode.
- rules_btc.h: This module controls the rendering of all pages of the rules section of "Beat the Clock" mode .
- rules_menu.h: This module controls the rules section and directs to rules section to all the modes.
- rules_replication.h: This module controls the rendering of all pages of the rules section of "Replication" mode.
- sdl_handler.h: This module handles all the functions directly related to SDL. Also provides abstraction of texture and timer functions.
- string.h: This module contains the necessary functions related to string.
- table.h: This module controls the state of the table.
- text_box.h: This module controls the textbox for prompting user input for names and password in login menu.
- time.h: This module controls timer and contains some necessary functions about time.
- user.h: This module controls the state of user data and controls user registration and authentication.
- vec.h: This module contains necessary functions about vectors.
- window_handler.h: This module controls the current window of the game and refers to expected modules and its functions.

5. Team Member Responsibilities:

- 8ball.h : Munawar Shakil Muhit
- 9ball.h : Munawar Shakil Muhit

- aimline.h: Munawar Shakil Muhit
- ball.h: Munawar Shakil Muhit
- btc_mode.h: Munawar Shakil Muhit
- button.h: Hasan Nur Towha
- circle.h: Munawar Shakil Muhit
- consts.h: Hasan Nur Towha
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- vec.h: Munawar Shakil Muhit
- window_handler.h: Hasan Nur Towha

6. **Platform, Library and Tools:**

As of now, this game is available for the Linux Operating System. It is developed with the help of SDL library, and its extension libraries: SDL_Image, SDL_Ttf and SDL_Mixer.

7. **Limitations:**

While the project seems to be quite well developed, some features were missed. These include:

   a. The game lacks animations and smooth transitions between windows.
   b. The game is available only on the Linux Operating System.

8. Conclusions:

At the very beginning, the game was supposed to feature only one mode, the "8 Ball Pool" as evident from its initial name which was eventually rethought and revised. During the development of "Billiard++", significant challenges were faced especially while implementing the collisions of ball, ball movement and aim line. Besides, a lot of planning and efforts were invested for designing the UI and implementing the login utility.

This project provided us with an opportunity to apply the theoretical concepts that we have learnt in our programming course and combine them with our own imagination and thinking. We also learnt about SDL libraries, its extensions as well as got valuable experience in graphics designing. Lastly, the project helped us to realise the importance of proper code structuring and writing clean code.

9. Future plan:

We are eager to introduce our game to the Android, iOS and Windows Platform. We also aspire to include online features in this game. We also intend to make further improvements to this game and provide the user with an even more refined experience.

10. Repositories:

GitHub Repository:https://github.com/Shakil-Muhit/Billiard-.git

YouTube Video: https://youtu.be/y6MpZOU-9Bc

11. References:

   a. SDL documentations (https://lazyfoo.net/tutorials/SDL/)
   b. Sound Effects (https://freesound.org/)